

The Specific-Factors Model

ECON 308 – International Economic Relations

Fall 2025

Problem Statement

Consider two countries, **Home** and **Foreign**, each producing two goods: **Manufactures (M)** and **Food (F)**.

- Labor (L) is mobile between sectors within each country but not internationally.
- Capital (K) is specific to manufacturing, and land (T) is specific to food.
- Production functions are:

$$Q_M = K^{1/2}L_M^{1/2}, \quad Q_F = T^{1/2}L_F^{1/2}, \quad L_M + L_F = L.$$

Endowments:

$$\text{Home: } K = 144, T = 64, L = 100 \quad \text{Foreign: } K^* = 64, T^* = 144, L^* = 100.$$

Assume competitive markets.

Part A. Autarky Equilibrium

Step 1: Marginal Products of Labor

Differentiate production functions:

$$MPL_M = \frac{\partial Q_M}{\partial L_M} = \frac{1}{2}K^{1/2}L_M^{-1/2}, \quad MPL_F = \frac{1}{2}T^{1/2}L_F^{-1/2}.$$

Interpretation: Both marginal products decline as labor increases in the sector (diminishing marginal returns).

Step 2: Wage Equalization Condition

Since labor is mobile, wages must be equal across sectors:

$$P_M \cdot MPL_M = P_F \cdot MPL_F.$$

This determines the division of labor between M and F .

Step 3: Home in Autarky

Let $P_M = 2$, $P_F = 1$. Home has $K^{1/2} = 12$, $T^{1/2} = 8$. Condition:

$$2 \cdot \frac{12}{\sqrt{L_M}} = \frac{8}{\sqrt{100 - L_M}}.$$

Simplify:

$$\frac{24}{\sqrt{L_M}} = \frac{8}{\sqrt{100 - L_M}}.$$

Square both sides:

$$\frac{576}{L_M} = \frac{64}{100 - L_M}.$$

Cross-multiply:

$$576(100 - L_M) = 64L_M \quad \Rightarrow \quad 57,600 - 576L_M = 64L_M.$$

$$57,600 = 640L_M \quad \Rightarrow \quad L_M = 90, \quad L_F = 10.$$

The wage:

$$w = P_M \cdot MPL_M = 2 \cdot \frac{12}{2\sqrt{90}} = \frac{12}{\sqrt{90}} \approx 1.265.$$

Explanation: Most labor is allocated to manufacturing since Home is capital-abundant.

Step 4: Foreign in Autarky

Foreign has $K^{*1/2} = 8$, $T^{*1/2} = 12$. Condition:

$$2 \cdot \frac{8}{\sqrt{L_M^*}} = \frac{12}{\sqrt{100 - L_M^*}}.$$

Simplify:

$$\frac{16}{\sqrt{L_M^*}} = \frac{12}{\sqrt{100 - L_M^*}}.$$

Square both sides:

$$\frac{256}{L_M^*} = \frac{144}{100 - L_M^*}.$$

Cross-multiply:

$$256(100 - L_M^*) = 144L_M^*.$$

$$25,600 - 256L_M^* = 144L_M^* \quad \Rightarrow \quad 25,600 = 400L_M^*.$$

$$L_M^* = 64, \quad L_F^* = 36.$$

The wage:

$$w^* = P_M \cdot MPL_M^* = 2 \cdot \frac{8}{2\sqrt{64}} = \frac{8}{8} = 1.$$

Explanation: Foreign allocates relatively more labor to food, since it is land-abundant.

Step 5: Autarky Relative Prices

- In Home, manufactures are relatively cheap (low P_M/P_F). - In Foreign, food is relatively cheap. *Implication:* Home will export M , Foreign will export F once trade is allowed.

Part B. Opportunity Costs and Comparative Advantage

Step 6: Opportunity Cost Formula

A reallocation of one unit of labor from F to M increases Q_M by MPL_M and reduces Q_F by MPL_F . Hence:

$$OC_M = \frac{MPL_F}{MPL_M}.$$

Step 7: Numerical Calculation at $L_M = L_F = 50$

- Home: $OC_M = \frac{8}{12} = \frac{2}{3} \approx 0.67$. - Foreign: $OC_M^* = \frac{12}{8} = 1.5$.

Step 8: Comparative Advantage

- Home's OC_M is lower $\Rightarrow$ comparative advantage in manufactures. - Foreign's OC_M^* is higher $\Rightarrow$ comparative advantage in food.

Conclusion: The pattern of trade is consistent with relative factor abundance.

Part C. Free Trade Equilibrium

Step 9: Home with World Price $P_M/P_F = 1.2$

Condition:

$$1.2 \cdot \frac{12}{\sqrt{L_M}} = \frac{8}{\sqrt{100 - L_M}}.$$

Simplify:

$$\frac{14.4}{\sqrt{L_M}} = \frac{8}{\sqrt{100 - L_M}}.$$

Square both sides:

$$\frac{207.36}{L_M} = \frac{64}{100 - L_M}.$$

$$207.36(100 - L_M) = 64L_M.$$

$$20,736 - 207.36L_M = 64L_M \Rightarrow 20,736 = 271.36L_M.$$

$$L_M \approx 76.4, L_F \approx 23.6.$$

Wage:

$$w = P_M \cdot MPL_M = 1.2 \cdot \frac{12}{2\sqrt{76.4}} \approx 0.825.$$

Interpretation: Compared to autarky, more labor shifts into manufacturing, but not as extreme as before because the world price is only moderately above 1.

Step 10: Foreign with World Price 1.2

Condition:

$$1.2 \cdot \frac{8}{\sqrt{L_M^*}} = \frac{12}{\sqrt{100 - L_M^*}}.$$

Simplify:

$$\frac{9.6}{\sqrt{L_M^*}} = \frac{12}{\sqrt{100 - L_M^*}}.$$

Square both sides:

$$\frac{92.16}{L_M^*} = \frac{144}{100 - L_M^*}.$$

$$92.16(100 - L_M^*) = 144L_M^*.$$

$$9,216 - 92.16L_M^* = 144L_M^* \Rightarrow 9,216 = 236.16L_M^*.$$

$$L_M^* \approx 39.0, \quad L_F^* \approx 61.0.$$

Wage:

$$w^* = 1.2 \cdot \frac{8}{2\sqrt{39}} \approx 0.769.$$

Step 11: Outputs

- Home:

$$Q_M = 12\sqrt{76.4} \approx 104.7, \quad Q_F = 8\sqrt{23.6} \approx 39.0.$$

- Foreign:

$$Q_M^* = 8\sqrt{39.0} \approx 49.9, \quad Q_F^* = 12\sqrt{61.0} \approx 93.6.$$

Step 12: Trade Pattern

- Home produces relatively more M , exports M , imports F . - Foreign produces relatively more F , exports F , imports M .

Part D. Distributional Effects of Trade

Step 13: Real Wage in Home

- In terms of M : $\frac{w}{P_M} = \frac{0.825}{1.2} \approx 0.69$ (lower than in autarky, workers lose relative to M). - In terms of F : $\frac{w}{P_F} = 0.825$ (higher than before, workers gain relative to F).

Workers face ambiguous welfare changes.

Step 14: Returns to Capital and Land in Home

- Capital: more labor in M raises its marginal product, capital owners gain. - Land: less labor in F lowers land productivity, landowners lose.

Step 15: Distribution in Foreign

- Workers: similar ambiguity as in Home. - Capital (specific to M): loses, since less labor allocated to manufacturing. - Land (specific to F): gains, as labor shifts into food.

Summary

- Home is capital-abundant $\Rightarrow$ comparative advantage in manufactures.
- Foreign is land-abundant $\Rightarrow$ comparative advantage in food.
- Trade benefits the specific factor used intensively in the export sector (capital in Home, land in Foreign).
- Trade hurts the specific factor used in the import sector (land in Home, capital in Foreign).
- Labor is mobile but experiences ambiguous welfare effects, depending on consumption patterns.